Coursework: Churn Prediction for FoodCorp

DUE DATE: See Moodle.

Overview

Foodcorp are once again asking for your expertise. Having considered the situation across their stores they have identified customer churn as an issue of concern that they would like to address.

They have recently hired a consulting company to provide a methodology for determining a data informed definition of churn and some initial analysis. This has been provided to you.

You have been tasked with continuing the development of the company's capacity to undertake churn prediction across all stores. The system they want will predict which customers will churn or not (binary prediction) and will be re-run regularly. Specifically, you have been asked to:

1) Interpret the initial work done on defining churn and finalise a definition of churn for the company. Based on this summarize the levels of churn. This initial work will be provided to you separately.

   This section evaluates:

   → Your ability to interpret existing analysis

   → Reason within a given business scenario in order to undertake analytics

2) Create and evaluate a churn prediction system so when they run the model (i.e. at different points in time) they can predict **active customers** who are likely to churn and take preventative action. They want **the feature engineering predominantly completed in SQL** and **the machine learning implemented in Python**[1]

   This section evaluates:

   → Your ability to undertake predictive analytics Python, for instance the sklearn framework

   → Your ability to identify and create relevant features within a given business domain primarily using SQL

   → Your ability to setup and evaluate temporal prediction problems in practice

   An **active customer** is defined as one who, at prediction time, is not already considered churned by your definition.

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[1] they have requested Python as they are expecting to (i.e. not part of this coursework) take your work and convert in into an automated system that makes predictions at regular intervals

3) Provide some insights into what differentiates people who churn vs. those that stay, including pen portraits.
   This section evaluations:
   → Your ability to interpret what is driving the predictions
   → Your ability to relate analytical outcomes to the business scenario

Dataset

- The dataset is a retail transactional database and will be an **individual dataset** (i.e. different for each of you). Full information will be provided, along with access instructions on moodle when the coursework is officially released.
- How to access your associated report (specific to your data) from the consultancy company will be made available on moodle when the coursework is officially released.
- The data has gone through basic data cleaning. What has been will be provided along with the data when it is released.
- As such, data cleaning is not an explicit task in this coursework. However, since the dataset is based on real world data errors may still exist. As in reality, as an analyst you are still expected to check the validity of your analytics as you go. If something doesn't look right and there is a possibility it is caused by data errors you need to check and potentially then document/correct.

What you need to submit

- A report. 8 pages maximum. See the report requirement section for the structure and to understand what is expected. The report is a technical report - see this document for tips and common mistakes people make when writing technical reports. It is written with your dissertation in mind, but almost everything still applies here. If you are unsure about technical writing make an appointment for a consultation with the Centre for English Language Education (CELE) - see Moodle for more information &/or ask in class.
- All code / notebooks / tableau files. At a minimum I expect:
  - A **commented** Databricks notebook containing the churn prediction component (see the tips/hints section regarding expectations on in-code comments)
- A supporting documentation document which should include at a minimum:
  - Instructions on how to run the churn model
  - Any additional cleaning you have undertaken (in most cases I expect this to be a statement that no extra cleaning was required)
  - A table detailing which python method/tableau file/… that was used to create each figure in the report.

How you need to submit

- All files must be zipped into one file
- Submission is electronically via Moodle.
- **DUE DATE: See Moodle.**

Report Requirements

Your report should contain the following. **Other than the executive summary**, page counts are only indicative and you can change the lengths as you see fit. Remember **8 pages is the maximum**, it is the quality of the report that is important not the length. Your report should contain both text and all graphs. Remember how (i.e. a statement/pointer to the code/tableau file) each figure was created should be included in a table outside of the report in the supporting document.

- A separate, short, AI (LLM/Agentic systems) declaration. See the AI policy for this coursework at the end of the document. This is required even if you did not use AI (LLM/Agentic systems).
- Executive Summary **[1 page]**
- Current levels of churn **[0.5 page]**
    - Including selected definition of churn
- Technical report on the developed churn prediction system. In all cases discuss things that were tried including the evaluation / reasoning as to why they were not finally included. You should at least includes a section on: **[4.5 pages]**
    - Features
        - List and justify which features you tried, which you kept/discarded and why
        - **Include a diagram** to aid your explanation of how you created your temporal features. This should make clear your windowing strategy and how your input and output features are temporally defined.
    - Prediction approach:
      Describe and justify the selected machine learning approach (set of steps including any preprocessing and the final predictive algorithm) you tried/used.
    - Evaluation
        - Include a subsection on what evaluation method you used. Use the terminology from the lectures.
    - Summary
- An insight report **[2 pages total - split into 2 separate sections]**
  This should detail the difference between churn and non-churn customers and be split into two parts. **These must be two separate sections.**
    - **Section 1:** A summary aimed at marketing.
      This should detail the insights in the context of the business and in non-technical language. These insights should be backed and linked to analytics detailed in the second part of the insight report.
        - These should be factual insights for marketing to use. Do not attempt to tell them how to do their job. I.e. DO NOT say "conduct a survey". Instead say: The data shows X. This indicates that a survey of Y is would provide Z. In this way you are not telling them something they should already know, but aiding their work with your analytics.
    - **May be included in either section:** Your pen portraits of churning and non-churned customers.
    - **Section 2:** A technical report detailing how these insights were derived.

Some notes/hints:

- Marks will be given for demonstrating understanding and critical thinking of the approaches attempted as well as for the final outcome. For instance, the report should not be of the form:

"I did this, I did that" but more focused and critical. I.e. "The final solution was [something]. This was compared against [something else] to investigate [another thing] with [this thing] showing improved performance likely due to [a good reason]."

- Marks will be given for clear documentation within your code and my ability to easily run it. This reflects the real world situation where the person who writes the code is typically not the one that ends up running it and/maintaining it in the long run. Note, however, that in code documentation should be concise. **This includes comments within your code.** Any function you write should have an in-code comment above it describing it and indicating that the parameters do. Comments should also be used every few lines to describe, in English, conceptually what is being done. I.e. by reading only the comments within the code (not the code) I should get a high level understanding of the approach that is being attempted.
- **Ensure you correctly use temporal evaluation as taught in this module.** TimeSeriesSplit and GridSearchCV (without manual index specification for temporal problems) are not appropriate temporal evaluation. Please see the lectures.

Marking Guidance

Marking will be by section. The provisional relative weights for these sections are included below, along with indications of what will be looked for.

Marks will be deducted if **the feature engineering is not predominantly completed in SQL** and/or **the machine learning is not implemented in Python.**

Also see this full marking rubric and marksheet.

Executive summary [10%]

→ Clarity and faithfulness as to the content being summarised

→ Appropriateness of the report for the intended audience

Initial analysis of churn rates and provision of churn definition [5%]

→ Demonstrated understanding and critical thinking in undertaking the task

→ Appropriateness/justification of approach

→ Correctness of implementation (if any, not expected)

Technical report on the developed churn prediction system: Feature selection, the prediction approach & evaluation [50%]

→ Demonstrated understanding and critical thinking in undertaking the task

→ Appropriateness/justification of approach (including the evaluation methodology)

→ Correctness of implementation

→ Clarity of report and documentation (including documentation of code via in-code comments)

Insight report part 1 [10%]

→ Clarity and faithfulness as to the content being summarised

→ Appropriateness of the report for the intended audience

Insight report part 2 [25%]

→ Demonstrated understanding and critical thinking in undertaking the task

→ Appropriateness/justification of approach

→ Correctness of implementation

→ Clarity of report and documentation